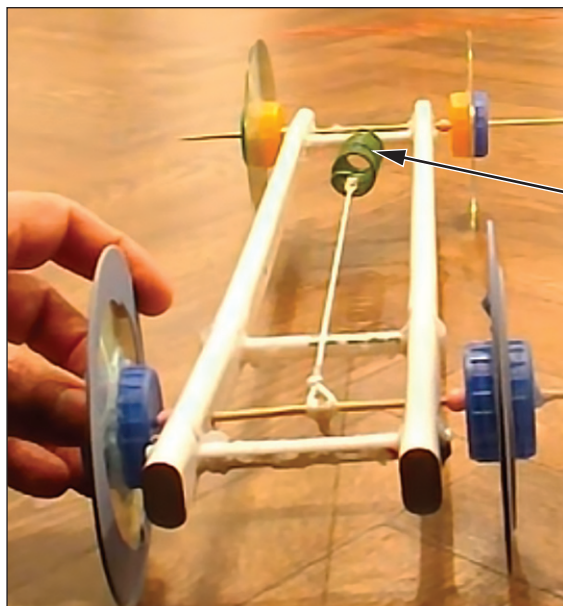
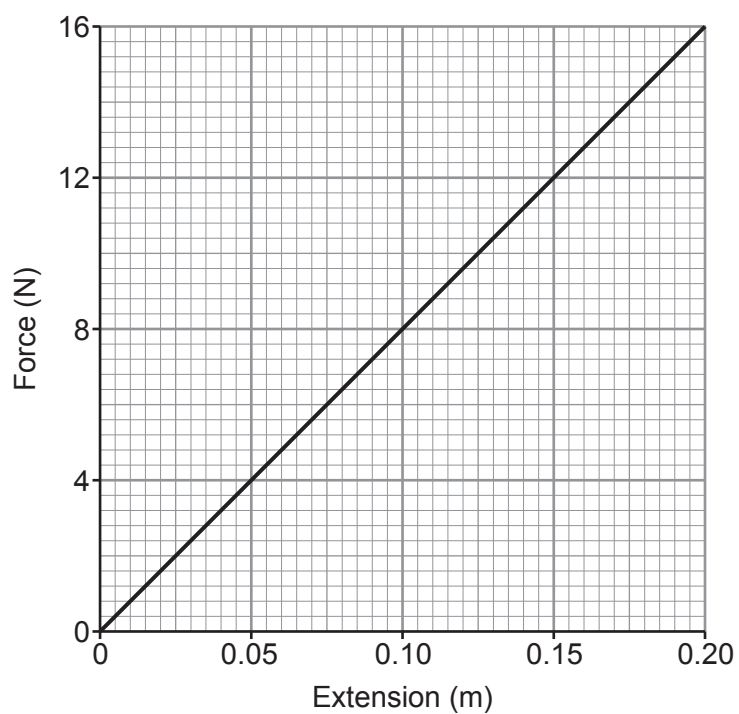


6. Some students make a car powered by a spring. The spring is stretched as the back wheels on the car are wound-up.



- (a) A force-extension graph for the spring used in the car is shown below.



- (i) Use the graph and an equation from page 2 to calculate the work done in stretching the spring by 0.15 m.

[2]

work done = J

- (ii) When the car is released the work done in stretching the spring is transferred to the car as kinetic energy.
Use your answer to part (i) and the equation:

$$\text{kinetic energy} = \frac{1}{2} \times \text{mass} \times \text{velocity}^2$$

to calculate the theoretical maximum velocity of the car if its mass is 0.075 kg.
Give the unit with your answer.

[4]

velocity =

unit =

- (iii) Explain why, in practice, the car will not reach the theoretical maximum velocity. [2]

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TURN OVER FOR THE REST OF THE QUESTION



- (b) The car's maximum kinetic energy is found to be 0.8 J.
A force brings it to rest in a distance of 1.5 m.

Use the equation:

$$\text{work done} = \text{force} \times \text{distance}$$

to determine the size of the force.

[3]

force = N

- (c) The spring constant of the spring used in the toy car is 80 N/m.
Maddie suggests that it would be better to use a spring with a spring constant of 160 N/m because for the same extension, double the energy would be stored in the spring.
Explain whether you agree.

[2]

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END OF PAPER

